

cpp-autodiff

A lightweight C++ implementation of **reverse-mode automatic differentiation** with a dynamic computational graph (inspired by MicroGrad/PyTorch internals).

Features

- Scalar **Value** node with:
 - **data**
 - **grad**
 - operation label (**op**)
 - parents (**prev**)
 - local backward function
- Supported ops:
 - **+**, **-**, *****, **/**, **pow**
- Topological graph traversal for **backward()**

Project Structure

- **src/value.hpp** — **Value** class + operator declarations
- **src/value.cpp** — operator implementations + **backward()**
- **src/main.cpp** — small demo

Build (g++)

From project root:

```
g++ -std=c++17 -I ./src src/value.cpp src/main.cpp -o demo
./demo
```

Build (CMake)

```
cmake -S . -B build
cmake --build build
./build/demo
```

Example

```
ValuePtr a = std::make_shared<Value>(2.0);
ValuePtr b = std::make_shared<Value>(-3.0);

ValuePtr c = a + b;
```

```
ValuePtr d = c * a;
```

```
d->backward();
```

Next Steps

- Add **Neuron**, **Layer**, **MLP**
- Add activation functions (**tanh**, **relu**)
- Add training loop and simple dataset
- Add tests